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Learning-Based UAV Coverage-Aware Path Planning in Large-scale Urban Environments

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Abstract. In our research, the coverage-aware path planning problem of unmanned aerial vehicles (UAVs) in large-scale urban environments is studied. The goal of our research is to train the intelligent UAV to autonomously cover the area with satisfactory communication quality and finally arrive at the objective place without colliding with numerous skyscrapers. Such a multi-objective path planning problem is regularly non-convex and thus hard to converge. To this end, a distributed deep reinforcement learning (DRL)-based approach called Distributed-RPPO is developed to address the problem. To be specific, firstly we formulate our proposed problem as a Markov decision process (MDP). Since communication connectivity is the core constraint, a proximal policy optimization (PPO)-based SIR prediction network is designed for guiding the UAV to cover the area with superior communication connectivity, using only the UAV's current location. Then we combine the PPO algorithm with the Long short-term memory (LSTM) structure to solve the basic navigation problem while only the historical interactive data is exploited for updating the parameter vector of the entire network. Simulation results demonstrate that our approach is capable of allowing the UAV to execute coverage-aware path planning tasks in large-scale urban environments. Furthermore, the learning efficiency is greatly improved compared to existing DRL-based approaches.

1. Introduction

Nowadays, the promising development of smart cities is providing convenience and efficiency for its citizens. Owing to unmanned aerial vehicles' (UAVs') flexibility and ease of deployment, they are anticipated to be significant components of smart cities, especially various internet of things (IoT) applications such as mobile edge computing, data harvesting, and UAV-enabled secure communication. Since these applications demand superior communication coverage around the UAV, cellular-connected UAVs [1][2] offer a feasible solution to this requirement, where UAVs are stacked into the next-generation cellular networks. Despite the attractive application prospect, there are still several crucial challenges to be overcome for the practical implementation of cellular-connected UAVs in urban environments. In particular, the aerial ubiquitous cellular coverage cannot be provided

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because the conventional cellular network is planned for terrestrial coverage [3]. Furthermore, in urban environments crowded with skyscrapers, the UAV will easily collide with these densely distributed obstacles, hence resulting in unaffordable damages involved to the security of the flight. Fortunately, these two challenges could be overcome by coverage-aware path planning, which directs the UAV in real time to cover areas with the optimal communication signal quality during flight. Therefore, in this paper, we concentrate on the UAV coverage-aware path planning problem in large-scale urban environments. This path planning problem is a multi-objective optimization problem, which is hard to solve for its non-convex characteristic. Moreover, with no prior knowledge about the environment, the UAV will get crashed easily during the initial training process, leading to low convergence speed. It is worth noting that there have been some previous works related to this problem.

To tackle this complicated path planning problem, [4]-[5] propose algorithms based on graph theory and convex optimization techniques. Despite the promising effects of the above methods, In practical urban environments, an accurate and tractable channel model is difficult to acquire, not to mention other restrictive constraints of the graph-based or map-based approaches like the poor generalization. In [6], a deep reinforcement learning (DRL)-based coverage-aware UAV path planning method is proposed by utilizing a dueling double deep Q-network (Dueling DDQN) with multi-step learning. Besides, an approach called Simultaneous Navigation and Radio Measurement (SNARM) is developed for generating optimal UAV moving policy to generate a satisfactory trajectory. [7] designs a decentralized deep reinforcement learning approach to solve the problem, and develop a signal-to-interference-plus-noise ratio (SINR)-prediction neural network to solve the coverage-aware problem. Nevertheless, these approaches do not take practical large-scale urban environments into consideration.

Other works emphasize on UAV navigation problem in large-scale urban environments, which is modeled as a partially observable Markov decision process (POMDP) and addressed by an actor-critic-based algorithm in [8]-[9]. However, the aforementioned methods can only fulfill the basic requirements for the UAV flying in the future city, avoiding collisions and reaching the destination. Besides, these methods could not be directly transformed into a more complicated path planning problem by adding the reward function involved with the communication quality.

In our research, to overcome the challenges of the coverage-aware path planning problem in large-scale urban environments, we propose a distributed recurrent policy proximal optimization (Distributed-RPPO) [10] algorithm leveraging the power of long short-term memory (LSTM) structure, enabling the UAV to cover the area possessing relatively better communication quality with a collision-free path to the destination. Our contributions are summarized as follows:

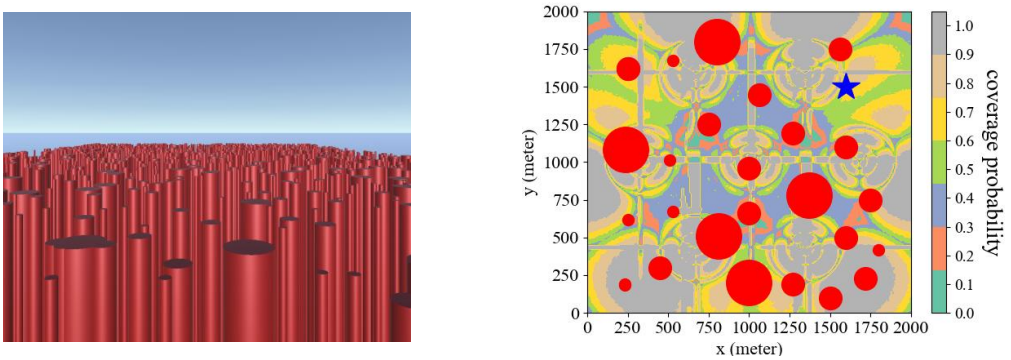
- We design a PPO-based signal-to-interference (SIR) prediction network to evaluate the communication quality of the possible positions in the next time step.
- We construct an RPPO-based navigation network to enable the UAV to avoid collisions and head for the destination.
- We design a reward function to comprehensively select a specific action based on the above two networks, allowing the UAV to complete our proposed navigation task.

2. Problem Description

In our research, we create a virtual urban environment as close to reality as possible, which is shown in Fig. 1. (a) as a three-dimensional (3D) view and in Fig. 1. (b) as a top view. In Fig. 1. (a), the densely distributed red cylinders denote numerous skyscrapers in the city, while in Fig. 1. (b) are denoted by red circles. Besides, the invisible communication environment is represented by the coverage map in Fig. 1. (b), where various colors denote different communication coverage probabilities.

Since we formulate the path panning problem as an MDP, the state space and the action space of the MDP need to be specified. Considering the navigation task, the surrounding aerial information is necessary. In this paper, we use ultrasonic rangefinders to detect the distance between the UAV and its surroundings. The rangefinder incorporates multiple sound waves, sent out from the UAV and

reflected while meeting an obstacle. To simulate the actual rangefinder, each sound wave has a detection range. In this way, the range finder reading $\mathcal{O}_t = [d_t^1, d_t^2, d_t^3, d_t^4, d_t^5, d_t^6, d_t^7, d_t^8]$ forms part of the state space as shown in Fig. 2. Due to the high relevance of the SIR level to the UAV's current position, another component of the state space is the UAV's current absolute position $\mathcal{P}_t = (x_t, y_t)$, which can be acquired by the GPS module of the UAV.



(a) (b)

Fig. 1. Simulated complicated urban environment. (a) 3D view. (b) top view.

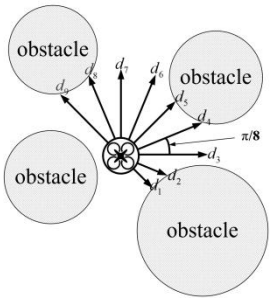


Fig. 2. The simulated ultrasonic rangefinder.

Thus, with the above definitions, the final state space is specified as $s_t = [d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8, \mathcal{P}_t] = [d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8, x_t, y_t]$. And the historical observation takes a setting that varies with time

$$h_t = (s_0, s_1, s_2, ..., s_t),$$
$$h_{t+1} \leftarrow h_t \cup s_t.$$

As for the action space, although the action space of the actual UAV is continuous, we utilize a discrete action space setting for simplicity $\mathcal{A}_t = [(1,0),(-1,0),(0,1),(0,-1)]$, corresponding to UAV's movement {right, left, front, back}.

Finally, the SIR model is defined as

$$\mathrm{SIR}_t = \frac{p_t^{b_t}}{\sum_{m \neq b_t} p_t^m}, \tag{1}$$

where b_t denotes the current number of base stations connected to the UAV, p_t^m denotes the received signal power at time step t by the UAV from cell m.

To quantitatively detect the SIR level, an outage probability function is defined as

$$P_{out}(\mathcal{P}_t, b_t) = \Pr(\mathrm{SIR}_t < \gamma_{th}), \tag{2}$$

where γ_{th} is a threshold value and $\Pr(\cdot)$ is the probability of the event. The formula can be transformed to empirical outage probability calculated by N SIR measurements:

$$\hat{P}_{out}(\mathcal{P}_t, b_t) = \frac{1}{N}\sum_{n=1}^{N} I(\mathcal{P}_t, b_t; \tilde{h}[t,n]) \tag{3}$$

where

$$I(\mathcal{P}_t, b_t; \tilde{h}[t,n]) = \begin{cases} 1, & \text{if SIR}(\mathcal{P}_t, b_t; \tilde{h}[t,n]) < \gamma \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

And $\tilde{h}[t,n]$ represents the small-scale fading.

3. Methodology

In this section, our proposed distributed DRL-based algorithm will be illustrated by two components, the SIR prediction network and the navigation network. An overview of the designed algorithm is shown in Fig. 3. And the specific architecture of each network is shown in Fig. 4 and Fig. 5. When the UAV is flying in the simulated large-scale city scenario, it is interacting with the complicated urban environment constantly. At each time step, the UAV acquires its current absolute location as the current state s_t and its historical state h_t , then the policy network of the SIR prediction network π_θ outputs the estimated outage probability distribution corresponding based on s_t , while the policy network of the navigation network π_ϖ outputs the selection probability distributions based on h_t . Afterward, the probability values output from both networks are superposed, then a sampling operation is executed to acquire the optimal action a_t :

$$a_t \sim \left[\pi_\theta(a_t \mid s_t) + \pi_\varpi(a_t \mid h_t)\right] \tag{5}$$

A discrete time step is deemed over when the action a_t is performed and the transition $(s_t, a_t, s_{t+1}, r_{SIR})$, $(h_t, a_t, h_{t+1}, r_{navi})$ are saved in the replay buffer. In the meantime, two networks are updated continually until we obtain the optimal policy to maximize the cumulative reward and a collision-free path with minimum outage duration.

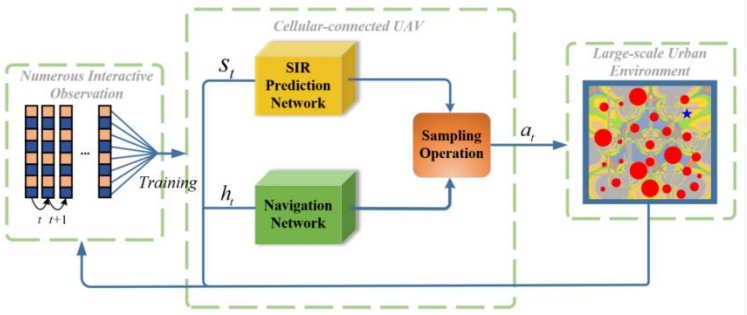


Fig. 3. An overview of the proposed distributed DRL algorithm.

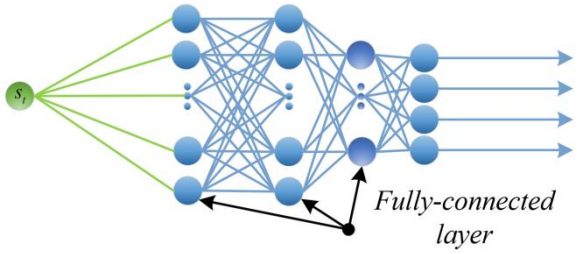


Fig. 4. The specific architecture of the value and policy network of the SIR prediction network

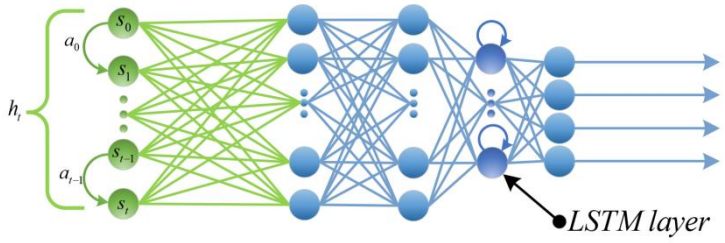


Fig. 5. The specific architecture of the value and policy network of the navigation network.

The SIR prediction network is designed based on an actor-critic architecture, utilizing the neural network function Q_ϕ as the value network and π_θ as the policy network, where ϕ and θ are parameter vectors. The two network takes exactly the same architecture except that in contrast to the policy network, the neuron number of the output layer of the value network is 1. The goal of the training of the SIR prediction network is to accurately predict the SIR corresponding to four possible actions. And it can be implemented by the stochastic gradient descent (SGD) of PPO

$$\theta_{k+1} = \arg\max_{\theta} \mathop{\mathbb{E}}_{s,a\sim\pi_{\theta_k}} \left[L(s,a,\theta_k,\theta) \right] \tag{6}$$

to maximize the loss function $L(h,a,\theta_k,\theta)$, which is specified as

$$L(s,a,\theta_k,\theta) = \min\left(\frac{\pi_\theta(a\,|\,s)}{\pi_{\theta_k}(a\,|\,s)} A^{\pi_{\theta_k}}(s,a),\ \ g\left(\varepsilon, A^{\pi_{\theta_k}}(s,a)\right) \right). \tag{7}$$

Where $A^{\pi_{\theta_k}}(s,a)$ denotes the advantage function and

$$g(\varepsilon,A) = \begin{cases} (1+\varepsilon)A & A \geq 0 \\ (1-\varepsilon)A & A < 0. \end{cases} \tag{8}$$

Note that how far the new policy can diverge from the old is determined by the relatively tiny hyper-parameter ε. Furthermore, the value network is updated by

$$\phi_{k+1} = \arg\min_{\phi} \frac{1}{|D_k|T} \sum_{\tau\in D_k} \sum_{t=0}^{T} \left(V_\phi(s_t) - \hat{R}_t \right)^2, \tag{9}$$

where D_k is the replay buffer incorporating multiple trajectories τ, and T is the total time steps of a complete trajectory, $V_\phi(\cdot)$ is the value function, and $\hat{R}_t$ is the reward-to-go.

The navigation network structure is almost identical to the SIR prediction network except that the third fully-connected layer is changed to an LSTM layer to deal with the historical information. Moreover, the updating process of this network is almost the same as the above operation, we only need to transform the input of the neural network to h_t.

Since a satisfactory strategy requires not only the proper network structure but also the appropriate reward design, in our research, we adopt a non-sparse reward setting. The non-sparse reward is distinguished for different networks.

Considering the prediction network, the main part of the reward is the empirical communication outage duration as defined in

$$r_{comm} = \mu_{comm} \hat{P}_{out}(p_t), \tag{10}$$

to prevent the UAV from lingering in the area with satisfactory communication connectivity, a move penalty r_{step} is designed. Thus the reward function of the SIR prediction network is set as:

$$r_{SIR} = r_{comm} + r_{step}, \tag{11}$$

As for the navigation network, firstly we need to stimulate the UAV to get closer to the destination and the r_{des} is defined as:

$$r_{des} = \nu d_{dist}, \tag{12}$$

where ν is a positive constant and d_{dist} denotes the difference of the distance to the destination between the next two time steps. To avoid obstacles in large-scale urban environments, we design an obstacle penalty as

$$r_{avoid} = -\alpha e^{-\beta d_{\min}}, \tag{13}$$

where α and β are positive constants. Note that a reward r_{free} is designed to guide the UAV to search for the open space, and the r_{step} is still crucial to prevent the UAV from spinning in the same space.
To summarize, the final reward for navigation is formulated as

$$r_{navi} = r_{avoid} + r_{des} + r_{free} + r_{step}. \tag{14}$$

As the different tasks are addressed by corresponding distributed networks, our proposed algorithm is named Distributed-RPPO.

4. Simulation results

As shown in Table 1, the experimental settings are as follows:

Table 1. Experimental Parameters.

Parameters	Value
Learning Rate	0.0001
Maximum number of episodes	6000
Replay buffer capacity	100000
Maximum steps per episode	200
N	1000
ε	0.2
μ_{comm}	20
r_{step}	-1
ν	0.5
α	3
β	2

In this paper, we utilize the RPPO algorithm and the deep recurrent Q-network (DRQN) algorithm as baselines to verify the convergence speed and effectiveness of the Distributed-RPPO. Fig. 6 demonstrates the moving average of return per episode obtained by different algorithms, it is observed that our proposed Distributed-RPPO outperforms other DRL algorithms in the convergence speed and the final convergence value. It can also be observed that the training results of the RPPO are still unstable until 5000 episodes and the DRQN totally loses effect in the complicated urban environments for coverage-aware tasks.

Table 2 demonstrates the statistical numerical results of 100 episodes. The low outage probability means that the communication connectivity between the UAV and the cellular infrastructure is greatly improved. Furthermore, the high success rate proves our algorithm's effectiveness. Fig. 7 validates that the Distributed-RPPO-based UAV always covers the area with optimal communication quality while avoiding collision with buildings.

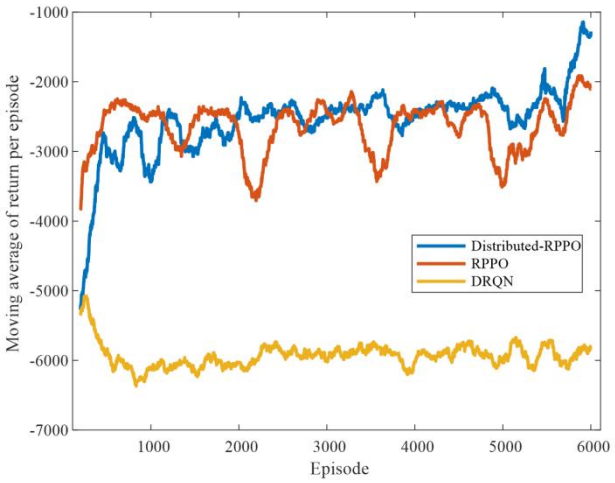


Fig. 6. The moving average of return per episode for the Distributed-RPPO, RPPO, and DRQN.

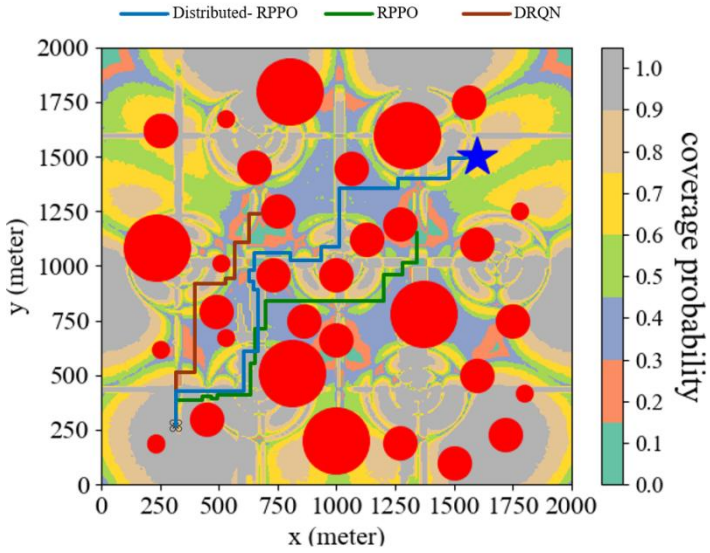


Fig. 7. A comparison between the paths generated by the Distributed-RPPO, RPPO, and DRQN.

Table 2. Numerical results.

	Success rate	Outage probability
Distributed-RPPO	92%	0.15
RPPO	73%	0.33
DRQN	15%	0.62

5. Conclusion

This paper proposes a DRL-based algorithm to address the coverage path planning problem in complicated large-scale urban environments. Simulation results demonstrate that the distributed architecture improves the algorithm's robustness and effectiveness. And the LSTM structure helps the UAV to capture more information for collision avoidance. The designed SIR prediction network is

capable of guiding the UAV to execute coverage-aware tasks. In future work, we are anticipated to extend the algorithm to a 3D communication coverage environment.

References

[1] Y. Zeng, Q. Wu and R. Zhang, "Accessing From the Sky: A Tutorial on UAV Communications for 5G and Beyond," Proc. IEEE, vol. 107, no. 12, pp. 2327-2375, Dec. 2019.

[2] Y. Zeng, J. Lyu and R. Zhang, “Cellular-Connected UAV: Potential, Challenges, and Promising Technologies,” IEEE Wirel. Commun., vol. 26, no. 1, pp. 120-127, Feb. 2019.

[3] 3GPP TR 36.873: “Study on 3D channel model for LTE”, V12.7.0, Dec., 2017.

[4] E. Bulut and I. Guvenc, “Trajectory optimization for cellular-connected UAVs with disconnectivity constraint,” in Proc. IEEE Int. Conf. Commun. (ICC), May 2018.

[5] S. Zhang and R. Zhang, “Trajectory optimization for cellular-connected UAV under outage duration constraint,” J. Commun. Netw., vol. 4, no. 4, pp. 55-71, Dec. 2019.

[6] Y. Zeng, X. Xu, S. Jin and R. Zhang, “Simultaneous navigation and radio mapping for cellular-connected UAV with deep reinforcement learning,” IEEE Trans. Wireless Commun., vol. 20, no. 7, pp. 4205-4220, Jul. 2021.

[7] X. Wang and M. C. Gursoy, “Learning-based UAV trajectory optimization with collision avoidance and connectivity constraints,” in Proc. IEEE Int. Conf. Commun. (ICC), 2021, pp. 1-6.

[8] C. Wang, J. Wang, X. Zhang and X. Zhang, “Autonomous navigation of UAV in large-scale unknown complex environment with deep reinforcement learning,” in IEEE Glob. Conf. Signal Inf. Process. (GlobalSIP), 2017, pp. 858-862.

[9] C. Wang, J. Wang, Y. Shen, and X. Zhang, “Autonomous navigation of UAVs in large-scale complex environments: a deep reinforcement learning approach,” IEEE Trans. Veh. Technol., vol. 68, no. 3, pp. 2124–2136, Mar. 20.

[10] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” 2017, available online at https://arxiv.org/abs/1707.06347.